

Pipeline AI

A CRM where the AI does the tedious half of selling: scoring every deal, drafting the follow-up, turning a call transcript into tasks, and answering questions about your pipeline in plain English. This guide covers every screen and every setting.

Contents

START HERE

- Running it
- Signing in
- The interface

EVERYDAY WORK

- Contacts
- Deals and stages
- Timeline and notes
- Tasks

THE AI FEATURES

- Lead scoring
- Draft follow-up

- Ask your CRM
- Meeting summaries
- Semantic search
- Duplicate detection
- Risk flagging

RUNNING THE SYSTEM

- Roles and access
- AI usage and cost
- Background jobs
- Configuration
- When AI is off
- Troubleshooting

Running it

Nothing needs installing beyond Node. The API starts its own database, its own job queue, and seeds a demo team, so the first run works with no configuration at all.

```
npm install
npm run dev    # app on :3000, API on :4000
```

The first start downloads a database binary and a small embedding model, then caches both. Later starts take a few seconds.

Demo data is not persistent by default

Without a database URL the app runs an in-memory database, so restarting clears everything and reseeds. That is deliberate for a trial. See [Configuration](#) to point it at a real database.

Signing in

Open `localhost:3000`. Three demo accounts exist, all with the password `password123`.

ACCOUNT	ROLE	SEES
admin@crm.dev	Admin	Every record, plus the duplicate queue and AI cost pages
ben@crm.dev	Member	Only the contacts and deals they own
cara@crm.dev	Member	Only the contacts and deals they own

Sign in as the admin to see the whole pipeline, or as a member to see how ownership scoping works. Sessions last a week and live in a cookie that JavaScript cannot read.

The interface

Three fixed areas, so nothing moves under you as you scroll.

- **The left rail** stays put while the page scrolls. A sparkle next to an item marks a screen where the AI does the work.
- **The status pill** in the header is the single source of truth for whether AI is live. Green with a model name means Claude is answering. Amber means the app is running on its built-in fallbacks. Hover it for the embedding model, vector store and queue in use.

- **The main column** holds the page. On the dashboard the right-hand rail follows you as you scroll the left column, so both halves stay useful.

The app follows your system light or dark setting. There is no theme switch to find.

Contacts

People, with the company they work for and the deals attached to them.

Creating and editing

New contact takes a name and optionally email, phone, company, tags and free-text profile notes. Only the name is required. Saving triggers a background duplicate check against everyone already in the system.

The list

Search matches name, email, company and tags at once. Every column header with an arrow sorts. The score column is the contact's strongest open deal, so sorting by it puts your best relationships on top.

The detail page

Deals for that person, the full timeline, their tasks, and a profile panel. Admins also see a banner when the contact has unreviewed duplicates.

Deleting is a cascade

Deleting a contact removes their deals, notes, tasks and meetings too. To collapse two records into one without losing anything, use the duplicate queue instead.

Deals and stages

A deal is a title, a contact, a value, a stage and an expected close date.

Lead Contacted Proposal Negotiation Won Lost

Stage colour warms as the deal advances, so the hue itself tells you roughly where something sits. Won and Lost step out of that ramp into their outcomes.

Change stage from the dropdown on the deal page or in the edit dialog. Every change writes a line to the timeline, resets the clock the stall detector watches, and queues a re-score.

Filtering the list

- Search covers deal titles and the associated contact and company.
- The stage dropdown narrows to one stage.
- **At risk only** shows just the flagged deals.
- Sort by score, value, close date, last activity or stage.

Timeline and notes

The timeline is the record of everything that happened, on both contacts and deals.

Log a note, call, email or meeting from the composer. `Ctrl + Enter` saves without reaching for the mouse. Three things then happen in the background, and none of them make you wait:

- 1 The note is classified for sentiment and a coloured label appears on it.
- 2 It is embedded so semantic search can find it by meaning.
- 3 The deal is re-scored and re-checked for risk.

Entries marked **System** are written by the app itself, for stage changes and merges. They carry no sentiment and do not count as engagement.

The "flagged" marker

A note containing instruction-like text ("ignore previous instructions...") is marked flagged. The content is kept and shown in full. The marker only tells you the text looked like an attempt to steer the AI, which the app treats strictly as data.

Tasks

Follow-ups, either typed by you or extracted from a meeting transcript. A task carries a title, an optional due date, and the deal or contact it belongs to.

Ticking one strikes it through immediately and the row collapses out of the list. Overdue dates turn red, and anything due within two days turns amber. The **Tasks** screen collects everything across your records; the **Due this week** card on the dashboard is the short version.

Tasks marked `from meeting` were created by the summarizer rather than by a person.

Lead scoring

Every deal carries a score from 0 to 100 estimating how likely it is to close. It is drawn as a ring, so you can read the pipeline without reading numbers.



70 to 100

Strong. Push to close.



40 to 69

Live, needs work.



Below 40

Cold or stalling.

Hover any gauge for the full breakdown. Six components add up, and each is shown with the input that produced it, so you can always see why a number moved.

COMPONENT	RANGE	WHAT MOVES IT
Stage prior	0 to 60	How far along the pipeline the deal is
Recency	0 to 12	Days since anyone touched it
Deal value	0 to 6	Size, on a log scale, so big deals do not dominate
Stage velocity	-15 to 6	Time in the current stage against that stage's threshold
Note sentiment	-12 to 12	Recent note sentiment, weighted toward the newest, plus a trend adjustment
Engagement	0 to 5	Human touches in the last 30 days

Won is always 100 and Lost always 0. The score recalculates whenever the deal or its notes change, and a nightly pass refreshes the recency component so untouched deals cool off on their own. **Recompute** on the deal page forces it immediately.

Why the model is arithmetic, not a prediction

The formula is deterministic so it can be tested, costs nothing to re-run, and can always be explained to whoever disagrees with it. The AI's contribution is the sentiment of each note, which feeds one component.

Draft follow-up

On any deal or contact, **Draft follow-up** writes the next email for you.

- 1 Optionally say what the email should do, for example "nudge on the revised pricing".
- 2 Pick a tone: professional, friendly or concise.

- 3 **Generate.** The model reads the contact, the deal and its stage, the last eight timeline entries, open tasks and the most recent meeting summary.
- 4 Edit the subject and body freely. It is an ordinary text box.
- 5 **Send & log to timeline**, or copy it, or open it in your mail client.

Nothing is ever sent automatically

The AI only ever fills the box. Sending is always a click you make, and the sent email is written to the timeline so the history stays complete.

Ask your CRM

Type a question about your pipeline and get a filtered table back, not a paragraph.

Questions it answers well:

- Show me deals over \$10k closing this month
- Which contacts have not been touched in 30 days?
- My at-risk deals, biggest first
- Open deals in negotiation with a score above 70
- Contacts at Northwind tagged enterprise

Every answer shows the query that actually ran as a row of chips, for example `value gt 10000`
`expectedCloseDate between 2026-09-01 and 2026-09-30`. You can always check the machine understood you.

What it will refuse

Anything that writes, deletes or sends. Anything needing analysis it cannot express as a filter, such as "why is this deal stuck". Anything unrelated to the CRM. It says so plainly rather than guessing.

The model never touches the database

It proposes a small typed filter. The server checks every field and operator against a fixed allowlist, resolves the dates, rejects anything shaped like an injection, then runs a read-only query with your own access scope applied. A member cannot widen their view by asking nicely.

Meeting summaries

Summarize meeting on a deal takes a call transcript, pasted or uploaded as `.txt` or `.vtt`, and returns four things.

OUTPUT	WHAT HAPPENS TO IT
Summary	Written to the deal timeline as a meeting entry
Action items	Created as tasks on the deal, with owner and due date where stated
Sentiment	Feeds the note sentiment component of the lead score
Next steps and topics	Shown on the meeting card for context

Processing happens in the background, so a long transcript never blocks the page. The card shows **Queued** then **Summarising** and fills in when done. Past meetings live on the **Meetings** tab of the deal.

Semantic search

Search your notes by meaning rather than by keyword. Searching **pricing pushback** surfaces "budget has been cut for this quarter" and "the quote is about 30% above their budget line", neither of which contains a word from the query.

Describe the situation you are trying to find. Each result shows how strong the match is, which deal or contact it belongs to, and the note's sentiment.

A badge tells you which mode ran. `Semantic match` means real meaning-based search. `Keyword fallback` means the embedding model or vector store was unavailable and the app fell back to plain text search, with the reason shown next to it.

Duplicate detection

Admin only.

New and edited contacts are checked automatically, and a full sweep runs nightly. Likely duplicates land in a review queue rather than being merged.

The matcher is built for the ways duplicates actually happen: a typo in an email, a mistyped domain, a nickname, a swapped name order, a reformatted phone number, a company written two ways. Two different people at the same company are not duplicates and are not flagged.

Reviewing

Each candidate shows both records side by side with the reasons they matched and, when AI is on, a verdict and confidence. Then you choose:

- **Keep this one** on the record you want to survive. Deals, notes, tasks and meetings move across, empty fields are filled from the other record, tags are combined, and the merge is logged on the timeline.
- **Not a duplicate** dismisses the pair so it stops coming back.

Merging never happens on its own

The queue is a suggestion. A person always picks the survivor. The retired record is hidden rather than deleted, so older links still resolve.

Risk flagging

Open deals are checked on every change and again by a daily scan. Four signals can fire:

SIGNAL	FIRES WHEN
Stalled	The deal has sat in its stage past that stage's threshold, 14 days early on and 21 later
Inactive	Nobody has touched it for longer than the inactivity window, 14 days by default
Sentiment negative	Recent notes are negative, or sentiment has dropped sharply against older notes
Closing soon, unready	The close date is within a week but the deal is still early stage

Flagged deals collect at the top of the dashboard, each with a plain-English reason and a suggested next step. When AI is on, Claude writes that explanation from the actual notes. When it is off, you get the rule-based version, which is shorter but just as specific.

A deal drops off the list as soon as the underlying signals clear. Won and Lost deals are never flagged.

Roles and access

	MEMBER	ADMIN
Contacts, deals, notes, tasks	Own records only	Everything
Ask your CRM, semantic search	Scoped to own records	Everything
Assign an owner	No	Yes
Duplicate review and merge	No	Yes
AI usage and cost	No	Yes
Create users, run jobs	No	Yes

Scoping is applied in the database query, not in the interface, so it holds on every route including the AI ones.

AI usage and cost

Admin only.

Every AI call is logged: tokens in and out, cache reads, latency, status and an estimated cost. The page breaks this down per feature over 1, 7, 30 or 90 days, with a live feed of recent calls.

Statuses worth knowing:

- ok a real model call, billed.
- cached served from the response cache. No tokens, no cost.
- fallback no provider configured, so the built-in path ran.
- error or timeout the call failed and the feature degraded.

The provider card shows the circuit breaker. After repeated failures it opens and calls fail fast instead of hanging, then recovers on its own. **Reset** clears it immediately.

Background jobs

Work that would make you wait happens off the request. Scoring, sentiment, embedding, summarizing, duplicate checks and risk assessment all run as jobs, and rapid edits to one record collapse into a single job.

SCHEDULED JOB	RUNS	DOES
Risk scan	Daily, 6am	Re-checks every open deal for risk signals
Rescore	Daily, 5am	Refreshes scores so untouched deals cool off
Duplicate sweep	Daily, 5.30am	Full pass over contacts

Admins can trigger any of the three by hand from the AI usage page.

Configuration

Copy `.env.example` to `.env`. Every setting has a working default; the ones below are the ones worth changing.

Turning on Claude

```
ANTHROPIC_API_KEY=sk-ant- ...
ANTHROPIC_MODEL=claude-opus-5
```

Restart, and the header pill turns green. Sentiment, drafts, summaries, question translation, risk explanations and duplicate verdicts are now written by the model.

Persistent data and durable jobs

```
docker compose up -d          # database + redis
MONGODB_URI=mongodb://localhost:27017/crm
REDIS_URL=redis://localhost:6379
npm run seed -w @crm/api      # add --reset to wipe
```

With a Redis URL set, jobs move to a durable queue with retries and real cron schedules. Without it they run in process, which is fine for a single machine but is lost on restart.

Embeddings and vector search

By default a small model runs inside the app and needs no key. Set `VOYAGE_API_KEY` or `OPENAI_API_KEY` to use a hosted one. For large note volumes, set `PINECONE_API_KEY` and `PINECONE_INDEX` to move vectors out of the database.

Tuning

SETTING	DEFAULT	EFFECT
RISK_INACTIVITY_DAYS	14	Silence before a deal counts as inactive
AI_TIMEOUT_MS	45000	How long an AI call may take before falling back
AI_RATE_LIMIT_PER_MINUTE	30	AI requests per user per minute
AI_CIRCUIT_FAILURES	4	Consecutive failures before the circuit opens
SMTP_URL	unset	Set to actually deliver email; otherwise drafts are logged only

When AI is off

No feature disappears without a provider, and none of them break a screen. Each degrades to something honest and says so in the interface.

FEATURE	WITHOUT A PROVIDER
Lead scoring	Sentiment comes from a keyword model. The score formula is unchanged.
Draft follow-up	A stage-appropriate template, labelled as one.
Ask your CRM	Built-in rules handle the common questions. Anything else says so.
Meeting summaries	Extracted summary and commitment-style action items.
Semantic search	Keyword search, with the reason shown.
Duplicates	Full matching, minus the written verdict.
Risk flagging	The same signals with a rule-based explanation.

The same applies when the provider is merely slow or erroring. A timeout falls back rather than hanging the page.

Troubleshooting

Everything logged me out

The API restarted while running the in-memory database, so it reseeded with new accounts. Sign back in. Set a database URL to stop this.

A score or risk flag looks stale

Both update on a background job a moment after the change. **Recompute** on the deal page forces it. If nothing moves, check the circuit breaker on the AI usage page.

Semantic search says keyword fallback

Either nothing has been embedded yet, or the embedding model is still loading on first run, or the vector store is unreachable. Hover the status pill to see which.

A meeting stayed queued

Open it and use **Retry**. If the provider was down, the retry will pick it up once it recovers.

My question was rejected

The rejection text says why. Usually the question asked for a write, or for an aggregate the filter language cannot express. Rephrase it as a filter over deals or contacts.